



Figure 3: Copying files to the office network

based authentication, which is inherently far more secure than password-based authentication. Here's how you set up this feature (details are very well illustrated at [8]):

1. Using ConnectBot, create a public key to be used with the remote (laptop) SSH server, with an algorithm such as 1024-bit RSA.
2. Copy the public key that's stored in the Android phone's memory.
3. Connect to the SSH server using your regular username and password.
4. In your home folder, create the folder `.ssh`, if it does not exist. In the `.ssh` folder, copy the public key file, renaming it to `authorized_keys`, and set its permissions to be readable and writable only by the owner (your login).
5. Test the connection by logging out of the SSH session, and reconnecting. If the key set-up is fine, you will not be prompted for the username or password.
6. Once this is working, disable password-based authentication for the SSH service: run `sudo gedit /etc/ssh/sshd_config`. In the editor window, search for a commented line, `#PasswordAuthentication yes`. Make a copy of this line just below it, delete the leading `#` to uncomment it, and change the `Yes` to `No`:

PasswordAuthentication no

After this, restart the SSH service (or reboot the laptop), and once again check that you can log in to the laptop from the phone. Disabling password authentication is good since even if you have a weak password for your regular login, scripted 'dictionary' attacks will no longer work.

Test the remote access!

Finally, you must test your set-up by actually accessing your home network machine from a different network, to ensure there are no unexpected connection problems. You could test from your office or coffee-shop Wi-Fi connection, or over your mobile service provider's data connection (3G or GPRS, for example).

Transferring files

The main reason for this set-up is to let you remotely access your laptop, and send the urgently required files to your office. There are multiple ways to achieve that; you could copy the files to your Android phone and mail them to the office, or transfer the files directly from your laptop to your office network. In our case, we used the secure copy utility [9] to copy the files from the laptop directly onto a publicly-accessible Linux SSH server in the office (the last two octets of the address are obscured on purpose):

```
Srg@laila: scp -P 22 violin root@69.89.xx.xx:/
```

Illustrations

Figure 1 shows the connection to the DynDNS hostname `laila.dyndns-server.com`, to the non-standard port 10102 we configured as the external port on the router, by using key-based authentication. Thus, no username or password is seen.

Figure 2 shows a successful login to the remote (laptop) SSH server.

Figure 3 shows that the `scp` command was used to copy a file to the office network (the publicly-accessible Linux server `69.89.xx.xx`). 

References

- [1] Popular website to find external IP address, <http://www.whatismyip.com/>
- [2] DynDNS Home Solutions, https://www.dyndns.com/about/home_solutions.html
- [3] DynDNS, detailed set-up instructions, <http://www.webupd8.org/2009/09/ssh-server-with-dynamic-ip-address-how.html>
- [4] Setting up a static IP address in Ubuntu, <http://linhost.info/2008/11/how-to-set-a-static-ip-on-ubuntu-810/>
- [5] OpenSSH, <https://help.ubuntu.com/7.04/server/C/openssh-server.html>
- [6] Uncomplicated firewall for Ubuntu, <https://help.ubuntu.com/community/UFW>
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- [8] Public Private Key encryption, <http://michaelchelen.net/2010/09/26/ssh-key-authentication-with-android-and-connectbot>
- [9] Secure Copy Utility, <http://manpages.ubuntu.com/manpages/lucid/man1/scp.1.html>

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